COMP 3133 – Lab Test – 2 (06%)

Submission Date and Time: 08 April 2026, 08:30 PM (Week – 13)

Objective: Create an Angular application to implement the following features.

Instructions:

1. **Create the Angular Application: (10 points)**
   - Create an Angular app named `studentid-lab-test2-comp3133`. ***e.g. 1012355-lab-test2-comp3133***
   - Initialize a GitHub repository and commit all your code to it.

2. **Host the Application: (10 points)**
   - Deploy your Angular application on a hosting platform (such as Vercel, Railway, Render, Docker, etc.).

3. **Create `MissionList` Component: (20 points)**
   - Create a component named `missionlist` to display a list of all SpaceX launches using the provided REST API endpoint.
     Endpoint: SpaceX Launches API
   - Display the following fields for each mission:
     - `flight_number`
     - `mission_name`
     - `launch_year`
     - `details`
     - `mission_patch_small`
     - `rocket`
       1. `rocket_name`
       2. `rocket_type`
     - `links`
       1. `mission_patch_small`
       2. `article_link`
       3. `Wikipedia`
       4. `video_link`

4. **Implement Search or Filter by Launch Year: (10 points)**
   - Create a `missionfilter` component that allows users to search/filter missions by their year of launch.
     Endpoint: SpaceX Launch Filter API

5. **Create `MissionDetails` Component: (20 points)**
   - Create a component named `missiondetails` to display detailed information about a selected mission from the `missionlist` component.
     Endpoint: SpaceX Mission Details API
   - Hint: Pass data to the `missiondetails` component using `@Input()` or use Route parameter.

6. **Create Service to Fetch Data: (10 points)**

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- Create a service to fetch data from the provided SpaceX REST API.

7. **Create Interface/Class for Data Structure: (10 points)**
   - Create an interface or class to define the structure of the data fetched from the SpaceX API.

8. **Use Angular Material for Design: (10 points)**
   - Use Angular Material components to enhance the design and user interface of the application.

Additional Optional Instructions (for using GraphQL API):

- **Optional**: If you prefer to use a GraphQL API instead of the REST API, you can integrate the SpaceX GraphQL API into your Angular application.
  - SpaceX GraphQL API: SpaceX GraphQL

  In this case, you will need to:

  - Set up a GraphQL client (e.g., Apollo Client) in your Angular project.
  - Modify the components and services to query the SpaceX GraphQL API instead of the REST API.
  - Update the data fetching and filtering logic to match the structure of the GraphQL API responses.

This option is **not mandatory**, but it may help you learn about integrating GraphQL with Angular.

Submission Requirements:

1. **Upload the Source Code**:
   - Upload a ZIP file containing your source code to Blackboard before the deadline.
2. **Provide GitHub Repository Link**:
   - In the comments section of your submission, include the link to your GitHub repository.
3. **Screenshots**:
   - Take screenshots of both pages of your application and upload them to D2L to show evidence of your work outside ZIP file.
4. **Hosting Link**:
   - Provide the link to your hosted application (on Render, Vercel, Railway, docker or another platform).

References for SpaceX API:

- SpaceX REST API Documentation
- **Optional**: SpaceX GraphQL API Documentation

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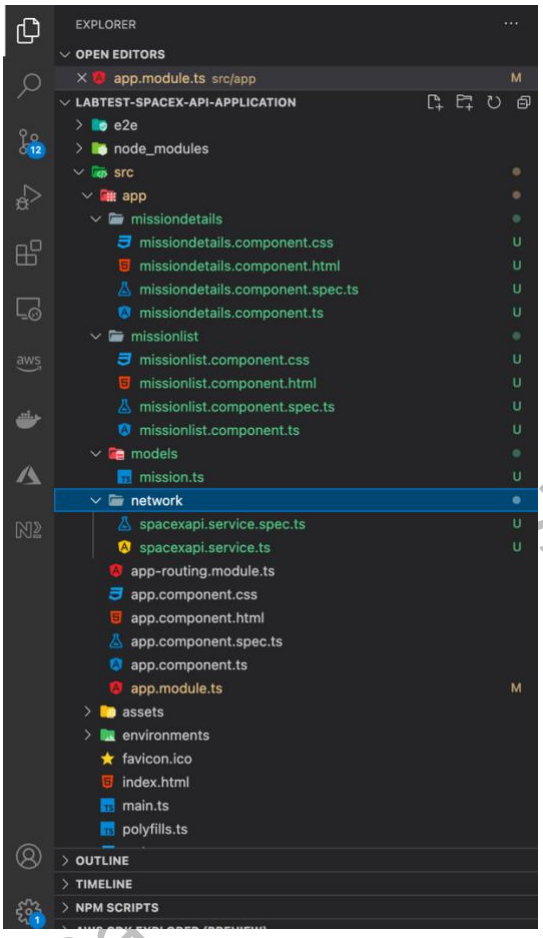
Folder Structure of application



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Sample Screen

Figure 1 Mission List

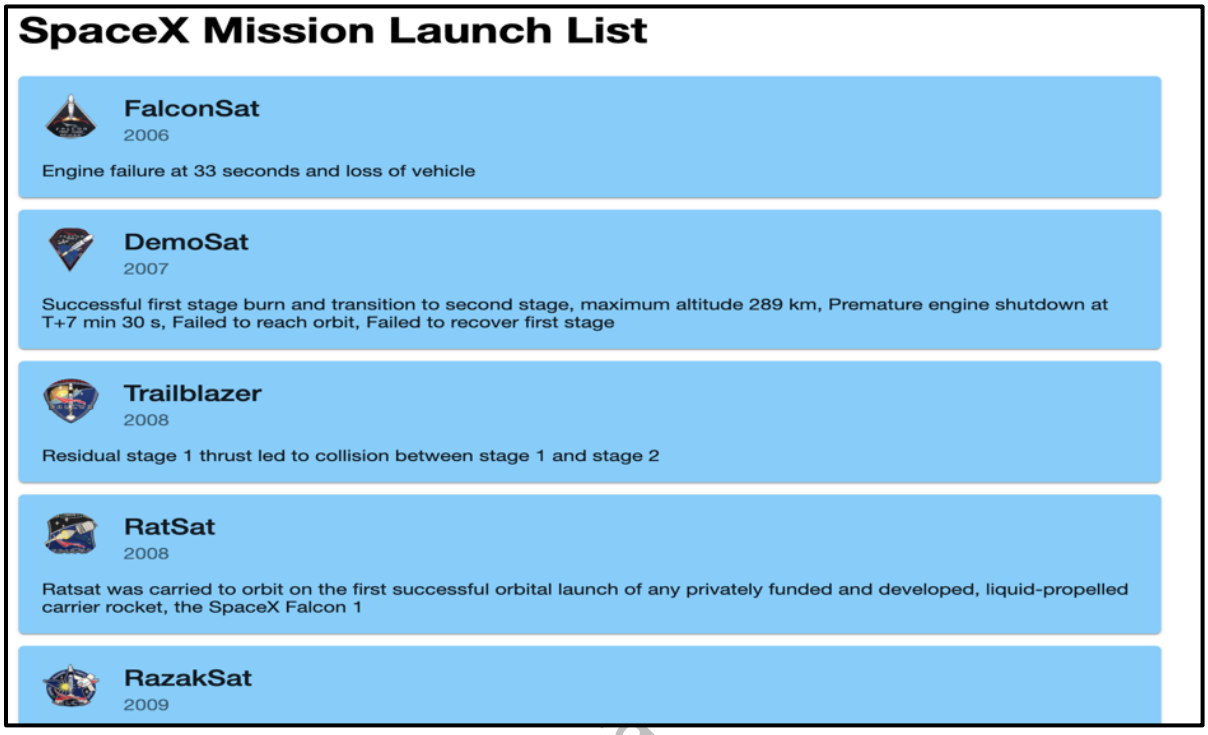

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Figure 2 Mission filter
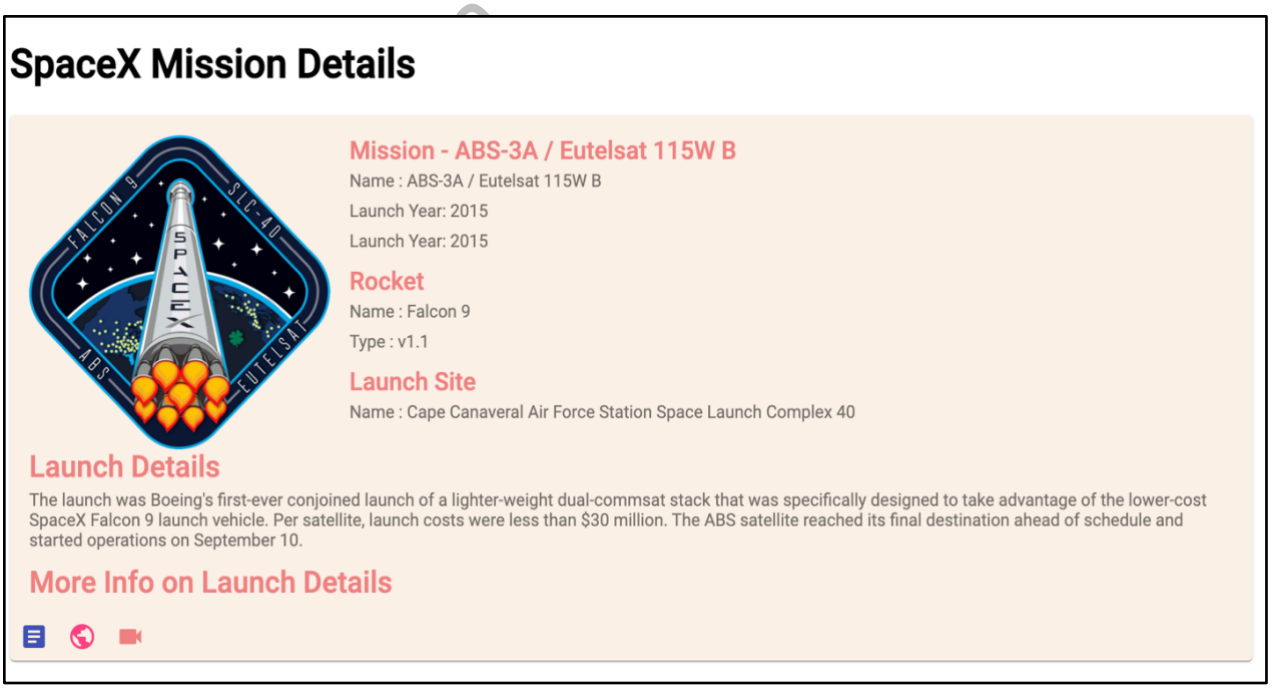
Figure 3 Mission Details